Quantification: 543-2
=====================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	8.5	8.5	2.7798	count
ring_length_r30_px	headline	369.583	363.5	87.4252	px
ring_length_r60_px	reported	1061.58	1079.5	212.423	px
owned_length_px	ownership-dependent	3482.71	3215.75	1346.8	px
edge_count	ownership-dependent	161	150	63.6025	count
mean_edge_length_px	ownership-dependent	21.6333	21.4151	2.0651	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.0419031	0.0416567	0.00587647	px^-1
ring_density_r60	normalised	0.047547	0.0480831	0.00496234	px^-1
roots_per_100px_perimeter	normalised	3.7455	3.6373	0.6692	per 100 px
ring_attached_length_r30_px	network v2	326.75	323.5	91.0236	px
ring_attached_length_r60_px	network v2	931.083	929.5	231.731	px
ring_length_w_r30_px	network v2	417.216	409.099	101.117	px
ring_length_w_r60_px	network v2	1213.1	1229.3	244.041	px
sholl_crossings_r10	network v2	9.75	9.5	2.9886	count
sholl_crossings_r20	network v2	12.8333	12	2.7907	count
sholl_crossings_r30	network v2	14.3333	14.5	3.7009	count
loop_count	loops, report only	19.0833	17	9.6621	count
loop_edge_share	loops, report only	0.438	0.4396	0.0984	share
ring_width_mean_r30_px	width	7.701	7.7684	0.3741	px

Per field
----

measure	value	units
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	44105	px
canalicular_length_density_per_px	0.0430611	px^-1
junction_count	1118	count
junction_density_per_px2	0.00109154	px^-2
lacuna_count	12	count

lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	150	count
mean_component_length_px	294.033	px
median_component_length_px	20	px
field_length_density_w_per_px	0.04904	px^-1
field_density_without_flagged_per_px	0.0430611	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	465	px
unattached_component_count	26	count
canalicular_length_density_attached_per_px	0.0430611	px^-1
field_length_density_with_unattached_per_px	0.0435152	px^-1
loop_count	275	count
loop_edge_share	0.4908	share
edge_count	2231	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.024	px
width_median_px	8	px
width_p10_px	5.6569	px
width_p90_px	10.198	px
width_pixels_used	44105	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	449	1329	4276.31	203	21.0656	7.9755
2	11	561	1428	5687.42	273	20.833	7.2256
3	6	338	925	2053.68	89	23.0751	7.8081
4	9	381	709	1205.83	70	17.2262	7.1387
5	5	279	853	3453.51	161	21.4504	8.0304
6	10	354	1023	2330.4	109	21.3798	7.4527
7	4	219	842	2977.99	113	26.3539	8.156

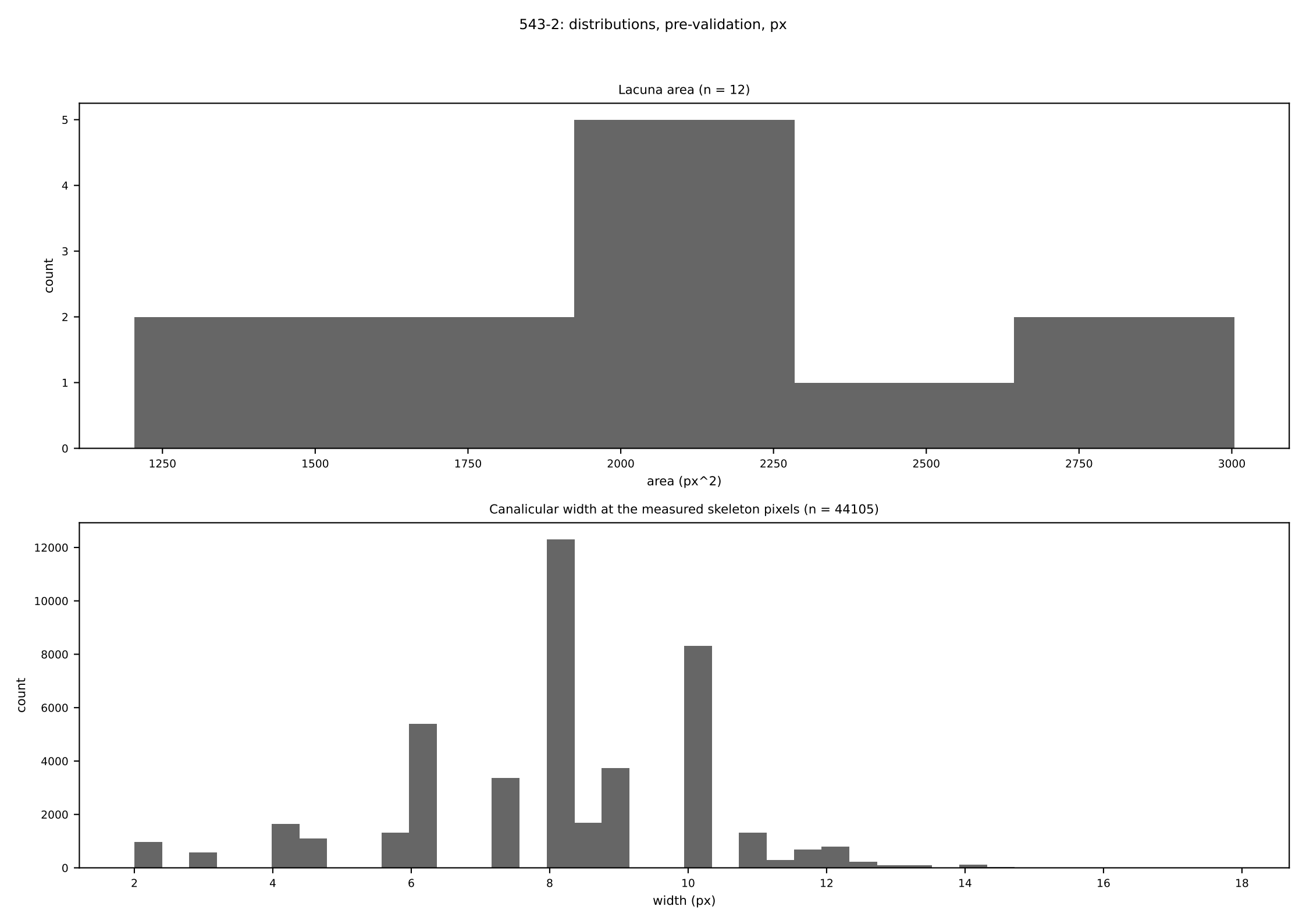
8	12	425	1180	2941.22	135	21.7868	7.8171
9	11	373	1172	4403.73	201	21.9091	7.7288
10	8	333	1136	5377.4	252	21.3389	8.2704
11	8	408	1177	4218.45	187	22.5585	7.5027
12	6	315	965	2866.53	139	20.6225	7.3062

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	244.492	9510	23791	1	0.9843	0.0472135	0.0558615	4.9081
2	295.12	10519	26271	1	1	0.0533321	0.0543565	3.7273
3	172.894	7548	18977	1	1	0.0447801	0.0487432	3.4703
4	232.551	8756	18925	0.9575	0.7928	0.043513	0.0374637	3.8701
5	154.267	7010	17818	1	1	0.0398003	0.0478729	3.2411
6	231.764	9080	22988	1	1	0.0389868	0.0445015	4.3147
7	155.924	7017	19168	1	1	0.0312099	0.0439274	2.5654
8	254.25	9429	23720	1	1	0.0450737	0.0497471	4.7198
9	277.078	10094	25807	1	1	0.0369526	0.045414	3.97
10	227.622	8939	23523	1	1	0.0372525	0.0482932	3.5146
11	225.522	8872	23390	1	1	0.0459874	0.0503207	3.5473
12	193.723	8132	21901	1	1	0.0387359	0.0440619	3.0972

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	420	1261	511.546	1532.05	12	16	16
2	506	1323	636.801	1628.47	15	19	20
3	294	752	381.463	1062.48	9	12	14
4	355	646	428.534	792.955	11	13	8
5	200	589	308.48	987.749	5	10	12
6	335	1001	403.706	1172.76	10	11	11
7	191	749	245.752	961.495	6	9	10
8	408	1032	489.304	1348.5	13	12	17
9	297	837	414.492	1346	8	15	14
10	252	1084	377.735	1285.83	7	11	15
11	351	1041	456.605	1330.26	12	14	20
12	312	858	352.179	1108.62	9	12	15


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 44105)
count
width (px)
0
2000
4000
6000
8000
10000
12000
2
4
6
8
10
12
14
16
18